

Article

Identification and Pathogenicity of *Botryosphaeriaceae*, *Colletotrichum*, and *Phytophthora* Species Associated with Avocado Diseases in Italy

Benedetto T. Linaldeddu ^{1,*} , Carlo Bregant ¹ , Jacopo Muscas ¹, Lucia Maddau ², Laura Vecchio ³ ,
Giancarlo Polizzi ³  and Dalia Aiello ³ 

¹ Dipartimento Territorio e Sistemi Agro-Forestali, Università degli Studi di Padova, 35020 Legnaro, Italy; carlo.bregant@unipd.it (C.B.); jacopo.muscas@studenti.unipd.it (J.M.)

² Dipartimento di Agraria, Università degli Studi di Sassari, Viale Italia, 39, 07100 Sassari, Italy; lmaddau@uniss.it

³ Dipartimento di Agricoltura, Alimentazione e Ambiente, Università degli Studi di Catania, Via Santa Sofia 100, 95123 Catania, Italy; laura.vecchio@phd.unict.it (L.V.); gpolizzi@unict.it (G.P.); dalia.aiello@unict.it (D.A.)

* Correspondence: benedetto.linaldeddu@unipd.it

Abstract

With the rapid expansion of avocado cultivation in southern Italy, growers have had to deal with the emergence of new diseases often caused by invasive and polyphagous pathogens responsible for leaf spot, branch cankers, dieback and fruit and root rot. Given the severity of these emerging diseases, a study was conducted in the main avocado growing areas in Sardinia and Sicily (Italy) to isolate and characterize the causal agents. Specifically, a total of 430 symptomatic leaf, fruit, branch, stem and root samples were collected and examined. Isolations performed on both non-selective and selective growth media yielded 22 species (fungi and oomycetes) belonging to the genera *Botryosphaeria*, *Colletotrichum*, *Diplodia*, *Dothiorella*, *Macrophomina*, *Neofusicoccum* and *Phytophthora*, including 14 new host-pathogen records in Italy. Notably, *Neofusicoccum australe*, *Phytophthora cinnamomi* and *Phytophthora palmivora* emerged as the main pathogens involved in the emerging avocado diseases. The identified pathogens were often isolated simultaneously from the same plants, which exhibited a complex of symptoms. Pathogenicity bioassays have helped to clarify the differences in aggressiveness among the different species and their specificity towards the different plant organs. The results achieved suggest that avocado orchards' productivity and profitability is threatened by a plethora of unrelated pathogens whose control represents a major challenge for the success of this crop in Italy.

Keywords: *Botryosphaeriaceae*; oomycetes; co-infections; holistic diagnosis



Academic Editor: David Guest

Received: 4 April 2026

Revised: 4 May 2026

Accepted: 5 May 2026

Published: 10 May 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Avocado is one of the most profitable tropical fruit crops worldwide, with approximately 700,000 ha cultivated mainly in Mexico, Colombia, Peru, the Dominican Republic and Kenya [1–3]. In Italy, avocado orchards are expanding in southern regions with an overall growing area of about 1.000 ha, mainly located in Sicily, followed by Calabria, Apulia and Sardinia [4]. In recent years, several emerging diseases were observed in southern Italian regions, with declining trees leading to substantial losses in productivity, fruit quality and profitability [5,6].

Globally, avocado production is impaired by several diseases, including fruit and leaf anthracnose caused by *Colletotrichum* spp. [7], branch cankers and dieback due to *Botryosphaeriaceae* species [8], and root rot caused by *Phytophthora* spp. [9].

Over twenty *Colletotrichum* species belonging chiefly to the “*acutatum*”, “*boninense*” and “*gloeosporioides*” species complex have been recognized as important avocado anthracnose agents worldwide [10]. *Colletotrichum* species can infect fruits, leaves, flowers and shoots, leading to substantial yield losses [11–13]. In Italy, *Colletotrichum fioriniae*, *C. fructicola* and *C. gloeosporioides* have been associated with postharvest fruit rot and stem-end rot diseases [6,14]. Moreover, recently *C. gloeosporioides* and *C. perseae* were reported as being responsible for necrotic lesions and dieback of young plants’ cv. Hass after grafting in a nursery [15].

In avocado, *Botryosphaeriaceae* are chiefly involved on twig, branch and stem cankers characterized by necrotic and sunken bark lesions, often associated with whitish and dark reddish exudates [5]. Internally, the wood becomes light-brown and gradually takes on the typical V-shaped appearance that is visible in a cross-section [16]. Bark and xylem necrotic lesions cause an interruption in the flow of water and nutrients, which leads to wilting and dieback of twigs and branches. In Italy, branch cankers caused by *Botryosphaeriaceae*, and in particular by *Neofusicoccum parvum*, have emerged as a significant threat to avocado production [5,6,16].

Reports of avocado root rot caused by *Phytophthora* spp. are increasing worldwide, with particular emphasis on the invasive species *P. cinnamomi* which is recognized globally as the main limiting factor to the avocado industry [9,17–19]. However, other *Phytophthora* species are also emerging as a new threat to avocado production, including *P. menzei* in California and Mexico [20,21], *P. heveae* in Colombia [22], *P. nicotianae* in Thailand [23], *P. niederhauserii* in the Canary Islands [24] and *P. palmivora* in Cuba [25]. Despite *Phytophthora* root rot representing the largest threat to the avocado industry in both traditional and emerging producing countries, to date, no exhaustive studies have been conducted in Italy to determine the presence and impact of these pathogens in avocado orchards, although reports from growers on the presence of typical *Phytophthora* symptoms such as root rot are progressively increasing in several producing areas.

Therefore, considering the growing impact of *Botryosphaeriaceae*, *Colletotrichum* and *Phytophthora* species on avocado production worldwide, the rapid expansion of avocado cultivation in southern Italy and the growing reports of new diseases characterized by a complex of symptoms in Italian avocado nurseries and orchards, the main aim of this work was to isolate and identify the main causal agents, as well as to evaluate the potential co-infection by multiple pathogens at the site and plant levels.

2. Materials and Methods

2.1. Field Surveys and Sampling

From summer 2022 to autumn 2024, extensive surveys were conducted in three nurseries and three orchards in Sicily and Sardinia (Italy), respectively (Table 1). Avocado plants of different age in both nurseries and orchards were checked for the presence of symptoms in the fruits (necrotic lesion and rot), leaves (necrosis), stem and branches (cankers, dieback and exudates) and at the collar and root system (exudates, necrosis and rot causing loss of fine roots). A total of 430 symptomatic avocado samples from 115 plants were randomly collected, placed in sterile plastic bags, labeled and then transferred to a laboratory for diagnostic analysis (Table 1).

Table 1. Details of sampling sites and numbers of avocado samples collected during field surveys.

Site	Latitude	Longitude	Cultivar	Condition	Age	Root	Collar	Stem/ Branch	Leaves	Fruits
1 §	37.215090	15.034110	Day	Potted plant	1	21	-	21	63	-
2 §	37.744896	15.195599	Hass/Fairchild	Potted plant	2	6	-	6	18	-
3 §	37.744896	15.195599	Hass/Zutano	Potted plant	5	9	-	-	-	-
4 °	39.433330	8.850343	Hass/Bacon	Orchard	9	25	4	90	25	66
5 °	39.432591	8.850415	Hass/Bacon	Orchard	9	5	1	25	10	5
6 °	39.433916	8.848711	Hass/Bacon	Orchard	3	20	-	5	5	-
Total						86	5	147	121	71

§ Sicily and ° Sardinia.

To estimate the disease incidence (DI), 36 avocado trees along three liner transects (12 per transect) were monitored during summer 2023 at site 4 in Sardinia. Given the uniformity of the symptoms in the plants, the transects were identified *at random* over the entire surface area of approximately 1 hectare. The disease incidence was evaluated as the number of symptomatic trees among the total number of avocado trees monitored. On the same trees, the occurrence of symptomatic fruits and premature fruit drop (carpoptosis) was also evaluated in August 2023 as the percentage of symptomatic fruits and fruit drop compared to the total number of fruits per plant.

2.2. Fungal and *Phytophthora* Isolations

In the laboratory, leaf, fruit, branch, stem (5 × 5 cm fragment including outer and inner bark) and root samples were surface disinfected with 70% ethanol for 30 s, rinsed in sterile distilled water, and then air-dried in an aseptic condition. From each sample, ten fragments (5 × 5 mm) aseptically cut from the margin of the necrotic lesion were placed onto both potato dextrose agar (PDA, Oxoid Ltd., Basingstoke, UK) and PDA supplemented with 100 mL/L of fresh carrot juice, 0.013 g/L of pimarinic acid and 0.05 g/L of hymexazol (PDA+) in 90 mm Petri dishes and then incubated at 20 °C in the dark for 7 days [26]. Hyphal tips from emerging colonies were sub-cultured onto PDA and carrot agar (CA) in 60 mm Petri dishes and incubated at 20 °C in the dark.

Indirect isolation from fine root and rhizosphere samples (100 g per tree), collected at a depth of 10 cm from 2 points 30 cm away from the trunk, were performed using the zoospore trap assay described by [27] with slight modifications. Each sample was placed in a Sedimentation Imhoff Cone into which 0.8 L of distilled water was added. After 24 h, fresh holm oak leaves taken from 2-month-old seedlings were placed on the clean water surface and used as baits to capture *Phytophthora* zoospores. The Imhoff cones were kept at 20 °C in laboratory conditions for 5 d. Oak leaves showing dark spots were cut into small fragments, air-dried on sterile filter paper in aseptic conditions and then placed in 90 mm Petri dishes containing PDA+ [26]. The dishes were incubated in the dark at 20 °C and examined daily. The pure cultures were stored as described above.

The surface and reverse colony appearance of fungal and fungal-like isolates was recorded on PDA and CA after 7 days at 20 °C in the dark. The biometric data of conidia (fungi) and sporangia (oomycetes) were recorded using a light microscope (Motic BA410E microscope) at ×400 magnification.

2.3. DNA Extraction, PCR Amplification and Phylogeny

Genomic DNA was extracted from aerial mycelium of 4-day-old cultures on PDA, using the Instagene Matrix kit (BioRad Laboratories, Hercules, CA, USA). The polymerase chain reaction (PCR) was performed in a total volume of 50 µL, using the PCRBIO Ultra

Polymerase mix (PCRBiosystems Ltd. London, UK) on a SimpliAmp™ Thermal Cycler (Life Technologies Holdings Pte Ltd., Singapore). The internal transcribed spacer (ITS) region of the rDNA was amplified and sequenced for all isolates, using the primer pairs ITS1 and ITS4 [28]. For the *Botryosphaeriaceae* species, a portion of the translation elongation factor 1 alpha gene (*tef1-α*) was also amplified and sequenced with the primers EF446f and EF1035r [29], whereas for *Colletotrichum* spp., a portion of the glyceraldehyde-3-phosphate dehydrogenase (*gapdh*) region using the primer pairs GDF1 and GDR1 [30]. For the ITS and *tef1-α* regions, the thermal parameters reported by Linaldeddu et al. [26] were used, whereas for *Colletotrichum* isolates, the PCR parameters were as follows: 95 °C for 4 min, followed by 35 cycles of denaturation at 95 °C for 30 s, annealing at 60 °C for 30 s and extension at 72 °C for 45 s, and a final elongation step at 72 °C for 5 min [31].

PCR products were purified using the Monarch™ PCR & DNA Cleanup Kit (New England Biolabs, Ipswich, MA, USA) according to the manufacturer's instructions. Specific amplicons of ITS, *tef1-α* and *gapdh* regions were sequenced by the BMR Genomics DNA sequencing service (www.bmr-genomics.it). The consensus sequences obtained by the FinchTV v1.4.0 (Geospiza, Inc.) software were compared with sequences of ex-type culture or representative isolates deposited in GenBank (<http://blast.ncbi.nlm.nih.gov>). New sequences were deposited and are available in GenBank (Table 2).

Table 2. Details of fungal and *Phytophthora* isolates included in the phylogenetic analyses. Ex-type cultures are given in bold and newly generated sequences are indicated in italics.

Species	Isolate Code	Host	GenBank Accession Code		
			ITS	<i>tef1-α</i>	<i>gapdh</i>
<i>Botryosphaeria agaves</i>	MFLUCC10-0051	<i>Agave</i> sp.	JX646790	JX646855	
<i>B. agaves</i>	MFLUCC11-0125	<i>Agave</i> sp.	JX646791	JX646856	
<i>B. corticis</i>	CBS 119047	<i>Vaccinium corymbosum</i>	NR111213	NR111213	
<i>B. corticis</i>	ATCC22927	<i>Vaccinium</i> sp.	DQ299247	EF614931	
<i>B. dothidea</i>	CBS 115476	<i>Prunus</i> sp.	KF766151	AY236898	-
<i>B. dothidea</i>	CFU 151	<i>Persea americana</i>	PX548330	PX584485	-
<i>Colletotrichum aenigma</i>	ICMP 18608	<i>P. americana</i>	JX010244	-	JX010044
<i>C. aenigma</i>	ICMP 18686	<i>Pyrus pyrifolia</i>	JX010243	-	JX009913
<i>C. alatae</i>	CBS 304.67	<i>Dioscorea alata</i>	JX010191	-	JX010011
<i>C. alatae</i>	ICMP 1791	<i>D. alata</i>	JX010190	-	JX009990
<i>C. boninense</i>	CBS 123755	<i>Crinum asiaticum</i>	JX010292	-	JX009905
<i>C. fructicola</i>	CBS 130416	<i>Coffea arabica</i>	JX010165	-	JX010033
<i>C. fructicola</i>	ICMP 12568	<i>P. americana</i>	JX010166	-	JX009946
<i>C. gloeosporioides</i>	CBS 112999	<i>Citrus sinensis</i>	JX010152	-	JX010056
<i>C. gloeosporioides</i>	CFU 76	<i>P. americana</i>	PX548331	-	PX584494
<i>C. hippeastri</i>	CBS 241.78	<i>Hippeastrum</i> sp.	JX010293	-	JX009932
<i>C. salsolae</i>	ICMP 19051	<i>Salsola tragus</i>	JX010242	-	JX009916
<i>C. salsolae</i>	CBS 119296	<i>Glycine max</i>	JX010241	-	JX009917
<i>C. siamense</i>	ICMP 18578	<i>C. arabica</i>	JX010171	-	JX009924
<i>C. siamense</i>	CFU 170	<i>P. americana</i>	PX548332	-	PX584495
<i>Diplodia fraxini</i>	CAD 001	<i>Fraxinus angustifolia</i>	KF307700	KF318747	-
<i>D. fraxini</i>	CAD 002	<i>F. angustifolia</i>	KF307701	KF318748	-
<i>D. mutila</i>	CBS 136014	<i>Populus alba</i>	KJ361837	KJ361829	-
<i>D. mutila</i>	CFU 165	<i>P. americana</i>	PX548333	PX584486	-
<i>D. sapinea</i>	CBS 393.84	<i>Pinus nigra</i>	DQ458895	DQ458880	-
<i>D. sapinea</i>	CBS 109943	<i>Pinus patula</i>	DQ458898	DQ458883	-
<i>D. seriata</i>	CBS 112555	<i>Vitis vinifera</i>	NR111151	AY573220	-
<i>D. seriata</i>	CFU 117	<i>P. americana</i>	PX548334	PX584487	-

Table 2. Cont.

Species	Isolate Code	Host	GenBank Accession Code		
			ITS	<i>tef1-α</i>	<i>gapdh</i>
<i>Dothiorella citricola</i>	CBS 130415	<i>Vachellia karroo</i>	MT587397	MT592109	-
<i>Do. citricola</i>	ICMP 16828	<i>Citrus sinensis</i>	EU673323	EU673290	-
<i>Do. mangifericola</i>	CBS 124727	<i>Mangifera indica</i>	KC898221	KC898204	-
<i>Do. mangifericola</i>	CBS 124726	<i>M. indica</i>	MT587407	MT592119	-
<i>Do. viticola</i>	CBS 117009	<i>V. vinifera</i>	AY905554	AY905559	-
<i>Do. viticola</i>	CFU 114	<i>P. americana</i>	PX548335	PX584488	-
<i>Macrophomina phaseolina</i>	CBS 205.47	<i>Phaseolus vulgaris</i>	KF951622	KF951997	-
<i>M. phaseolina</i>	CFU 113	<i>P. americana</i>	PX548336	PX584489	-
<i>M. tecta</i>	BRIP 70781	<i>Sorghum bicolor</i>	MW591684	MW592271	-
<i>M. tecta</i>	BRIP 71603	<i>S. bicolor</i>	MW591631	MW592218	-
<i>M. vaccinii</i>	CGMCC3.19504	<i>Vaccinium</i> sp.	MK687451	MK687426	-
<i>M. vaccinii</i>	CGMCC3.19503	<i>Vaccinium</i> sp.	MK687450	MK687427	-
<i>Neofusicoccum australe</i>	CBS 139662	<i>Acacia</i> sp.	AY339262	AY339270	-
<i>N. australe</i>	CFU 120	<i>P. americana</i>	PX548337	PX584490	-
<i>N. cryptoaustrale</i>	CBS 122813	<i>Eucalyptus</i> sp.	FJ752742	FJ752713	-
<i>N. cryptoaustrale</i>	CFU 107	<i>P. americana</i>	PX548338	PX584491	-
<i>N. luteum</i>	CBS 562.92	<i>Actinidia deliciosa</i>	KX464170	KX464690	-
<i>N. luteum</i>	CFU 118	<i>P. americana</i>	PX548339	PX584492	-
<i>N. parvum</i>	CMW 9081	<i>Populus nigra</i>	AY236943	AY236888	-
<i>N. parvum</i>	CFU 167	<i>P. americana</i>	PX548340	PX584493	-
<i>Phytophthora alpina</i>	CBS 146801	<i>Alnus viridis</i>	MT707332	-	-
<i>P. alpina</i>	OV2	<i>A. viridis</i>	MT707331	-	-
<i>Phytophthora bilorbang</i>	CBS 161653	<i>Rubus anglicandicans</i>	JQ256377	-	-
<i>P. bilorbang</i>	CB 489	<i>P. americana</i>	PX548341	-	-
<i>P. cinnamomi</i>	CBS 144.22	<i>Cinnamomum burmannii</i>	MG865473	-	-
<i>P. cinnamomi</i>	CB 102	<i>P. americana</i>	PX548342	-	-
<i>P. citricola</i>	CBS 221.88	<i>C. sinensis</i>	MG865475	-	-
<i>P. citricola</i>	CFU 174	<i>P. americana</i>	PX548343	-	-
<i>P. citrophthora</i>	P0479	<i>Citrus</i> sp.	MG865476	-	-
<i>P. citrophthora</i>	CFU 160	<i>P. americana</i>	PX548344	-	-
<i>P. gonapodyides</i>	P7050	<i>Eucalyptus obliqua</i>	MG865501	-	-
<i>P. gonapodyides</i>	CFU 139	<i>P. americana</i>	PX548345	-	-
<i>P. heterospora</i>	CBS 148034	<i>Olea europaea</i>	MT232393	-	-
<i>P. heterospora</i>	PH211	<i>O. europaea</i>	MT232394	-	-
<i>P. infestans</i>	CBS 147289	<i>Solanum tuberosum</i>	MG865512	-	-
<i>P. infestans</i>	P19941	<i>S. tuberosum</i>	MG865513	-	-
<i>P. insolita</i>	CBS 691.79	water	MG865515	-	-
<i>P. insolita</i>	WPC6703B561	soil	FJ801939	-	-
<i>P. inundata</i>	CBS 216.85	<i>Salix matsudana</i>	MG865516	-	-
<i>P. inundata</i>	VHS16836	<i>Xanthorrhoea preissii</i>	HQ012944	-	-
<i>P. irrigata</i>	MYA-4457	water	MH136914	-	-
<i>P. irrigata</i>	P16860	water	GU594791	-	-
<i>P. megakarya</i>	CBS 238.83	<i>Theobroma cacao</i>	MG865533	-	-
<i>P. megakarya</i>	IMI 202077	<i>T. cacao</i>	MG865534	-	-
<i>P. multivora</i>	CBS 124094	<i>Eucalyptus marginata</i>	MG865546	-	-
<i>P. multivora</i>	CB 125	<i>P. americana</i>	PX548346	-	-
<i>P. nicotianae</i>	P7661	<i>Nicotiana tabacum</i>	MG865550	-	-
<i>P. nicotianae</i>	CFU 181B	<i>P. americana</i>	PX548347	-	-
<i>P. niederhauserii</i>	CBS 149824	<i>Hedera helix</i>	MG865552	-	-
<i>P. niederhauserii</i>	CFU 175	<i>P. americana</i>	PX548348	-	-

Table 2. Cont.

Species	Isolate Code	Host	GenBank Accession Code		
			ITS	<i>tef1-α</i>	<i>gapdh</i>
<i>P. pachypleura</i>	IMI 502404	<i>Aucuba japonica</i>	KC855330	-	-
<i>P. pachypleura</i>	CBP 158	<i>C. sativa</i>	PQ571408	-	-
<i>P. palmivora</i>	P0633	<i>Areca catechu</i>	LC595875	-	-
<i>P. palmivora</i>	CFU 176	<i>P. americana</i>	PX548349	-	-
<i>P. plurivora</i>	CBS 124093	<i>Fagus sylvatica</i>	MG865568	-	-
<i>P. plurivora</i>	CFU 250	<i>P. americana</i>	PX548350	-	-
<i>P. polonica</i>	P131445	<i>Alnus glutinosa</i>	AB511828	-	-
<i>P. polonica</i>	CFU180	<i>P. americana</i>	PX548351	-	-
<i>P. sojae</i>	P3114	<i>Glycine max</i>	MG865587	-	-
<i>P. sojae</i>	P3248	<i>G. max</i>	MG865588	-	-

Sequences from 22 representative strains obtained in this study and 67 sequences of *Botryosphaeria*, *Colletotrichum*, *Diplodia*, *Dothiorella*, *Macrophomina*, *Neofusicoccum* and *Phytophthora* strains downloaded from GenBank including ex-type cultures were used in the phylogenetic analyses (Table 2). To confirm the identity of isolates at the species level, phylogenetic analyses were conducted first individually for each locus and then as combined analyses of ITS plus *tef1- α* for *Botryosphaeriaceae* and ITS plus *gapdh* for *Colletotrichum* species, respectively. Sequence alignments were performed with ClustalX version 1.83 [32], using the following parameters: pairwise alignment parameters (gap opening, 10; gap extension, 0.1) and multiple alignment parameters (gap opening, 10; gap extension, 0.2; transition weight, 0.5; delay divergent sequences, 25%). The alignments were checked and edited with BioEdit Alignment Editor version 7.2.5 [33]. Phylogenetic analyses were done with mega-X 10.1.8 [34]. All gaps were included in the analyses. The best model of DNA sequence evolution was determined automatically by the software. Maximum-likelihood (ML) analyses were performed with a neighbor-joining (NJ) starting tree generated by the software. A bootstrap analysis (1000 replicates) was used to estimate the robustness of nodes.

2.4. Pathogenicity Tests

The pathogenicity of the main species (isolated more frequently or for the first time from avocado) as well as their specificity towards the different plant organs, was evaluated using three pathogenicity bioassays conducted in a nursery and in a climate chamber. Specifically, five isolates belonging to the *Botryosphaeriaceae* family (*Diplodia mutila* CFU165; *Diplodia seriata* CFU 117; *Dothiorella viticola* CFU 114; *Neofusicoccum parvum* CFU 167; *Neofusicoccum australe* CFU120), one isolate of *Colletotrichum siamense* (CFU 170) and nine isolates of *Phytophthora* (*Phytophthora bilorbang* CB489; *Phytophthora cinnamomi* CB102; *Phytophthora citricola* CFU174; *Phytophthora citrophthora* CFU 160; *Phytophthora gonapodyides* CFU 139; *Phytophthora multivora* CB 125; *Phytophthora niederhauserii* CFU175; *Phytophthora palmivora* CFU 179; and *Phytophthora polonica* CFU 180) were used during the bioassays.

The first test was conducted on potted 1-year-old plants “Hass” grafted on “Day”. *Botryosphaeriaceae* isolates were previously grown on PDA amended with 100 mg L⁻¹ of streptomycin sulphate and incubated at 25 ± 1 °C for 7 days, whereas *Phytophthora* species were cultivated on corn meal agar (CMA, Oxoid™, Basingstoke, England) amended with 100 mg L⁻¹ of streptomycin sulphate and incubated at 25 ± 1 °C for 10 days. The stem was surface sterilized with a 70% ethanol solution, and the bark was removed using a sterile cork-borer (6 mm Ø). Then, the mycelial plug of each species was placed upside down onto each wound. The wounds were sealed with Parafilm® to prevent dehydration. A total of nine plants were inoculated with each isolate, while the control consisted of an

equal number of plants inoculated with sterile PDA. The temperature in the greenhouse ranged from 10 to 22 °C and the humidity from 70 to 80%. Plants were regularly watered and monitored monthly for symptom development. After three months, the length of the inner bark lesion extending both upward and downward from each inoculation point was measured and recorded.

The second test was performed using young detached avocado leaves “Hass”. Six leaves were used for each fungal and fungal-like isolate. The leaf surface was previously disinfected with 70% ethanol and injured in four points with a sterile needle, then a mycelial plug of each species, taken from the margin of colonies grown on PDA for 7 days, was placed upside down on each wound. The control consisted of leaves that were wounded and inoculated with a sterile PDA plug. Successively, the leaves were placed in a growth chamber at 25 ± 1 °C with a photoperiod of 16 h. After seven days, the size of the necrotic lesion was evaluated by measuring two perpendicular diameters at each inoculation point.

In the third test, artificial inoculations of asymptomatic fruits (cv. Bacon) were performed under controlled conditions. Twenty fruits per isolate were initially surface disinfected with 70% ethanol, then inoculated with a 5 mm-diameter mycelium PDA plug taken from an active 5-day-old colony on the stem end of fruits without injuries. The same number of fruits were inoculated with a sterile PDA plug and used as controls. The inoculated and control fruits were incubated in a climatic chamber at 25 °C in the dark with 70% relative humidity for 4 days. The two species most frequently isolated from fruits (*N. australe* and *P. palmivora*) were used in this assay. At the end of the experimental period, the occurrence of external and internal necrotic lesions was assessed, and lesion size was recorded.

Koch’s postulates were achieved for each bioassay by re-isolations from artificially inoculated tissues. For both the fungal and *Phytophthora* species, re-isolations were performed from symptomatic tissues cut around the margin of the necrotic lesions and placed onto PDA (fungi) or PDA+ (*Phytophthora*). Growing colonies were sub-cultured onto PDA, incubated in the dark at 20 °C for 7 days and identified based on colony appearance and DNA sequence data (ITS region).

2.5. Statistical Analysis

Results of the pathogenicity test were checked for normality, then subjected to analysis of variance (ANOVA). Significant differences among the mean values were evaluated using Fisher’s least significant differences multiple range test ($p = 0.05$) after one-way ANOVA using XLSTAT 2008 software (Addinsoft).

3. Results

3.1. Symptoms and Disease Incidence

A variety of symptoms in avocado leaves, fruits, twigs, branches, stems and roots were recorded in both the nursery and orchard (Figures 1–4). In the orchard, symptomatic avocado trees showed chlorosis, stunted growth, extensive shoot blight, dieback and sudden death (Figure 1a–c). On the leaves, two distinct disease symptoms were consistently observed: the first consisted of dark brown necrosis starting from the margin of the apical portion (DI: 86%) (Figure 1d,e); this type of necrosis progressively enlarged merging, leaving the branch partially defoliated. Less frequently, the presence of small leaf spots (anthracnose) randomly distributed across the entire leaf blade was observed, especially at the end of summer (DI: 3%); over time, the lesions progressively enlarged, merging into larger spots (Figure 1f).



Figure 1. Overview of foliar and canopy disease symptoms: chlorosis and stunted growth (a), extensive shoot blight (b), sudden death (c), apical and marginal leaf necrosis (d,e), and leaf anthracnose (f).



Figure 2. Overview of fruit disease symptoms: dark necrotic lesion affecting the exocarp and gradually migrating from the stylar region to the peduncle, causing a light brown rot (a,b); intense carpoptosis during the summer (c); stem end rot (d,e); and small slightly sunken brown light lesions (anthracnose) (f).



Figure 3. Overview of *Phytophthora*- (a–d) and *Botryosphaeriaceae* (e–k)-related disease symptoms: shoot blight (a), dark bark necrosis on branches (b), bleeding stem canker with a whitish and black exudation (c), root rot showing the characteristic “rat tail” symptoms (d), branch dieback (e,f), sunken canker (g), stem canker with hard whitish and black exudates (h), and a cross-section of sunken cankers highlighting the progressive expansion of the V-shaped necrotic sector (i–k).

Three main disease symptoms were observed on the fruits: the first was visible in late summer on unripe fruits as a dark necrotic lesion affecting the exocarp and gradually migrating towards the peduncle, causing a light brown rot (Figure 2a–c). In a few days, the infection determined an intense carpoptosis (DI: 19%). The second symptom (stem-end rot) was recognized chiefly in autumn and winter (ripe fruits) as a slightly sunken necrosis which migrated from the petiole towards the fruit (DI: 4%), whereas small slightly sunken brown light lesions (anthracnose) were detected on 3% of fruits (Figure 2f).

On shoots, branches and stem necrotic bark lesions (cankers) with and without a sunken appearance and sometimes with whitish hard exudates visible in the center of the canker were detected on 86% of the monitored trees (Figure 3). In cross-sections, cankers with a sunken appearance showed a wood reddish-brown necrosis that gradually took a typical V-shaped girdling of the branch (Figure 3i–k). In the canker without sunken bark lesions, the necrosis was confined to the inner bark.

Finally, 11% of trees in the liner plot showed typical root rot symptoms, with the loss of feeder roots and the presence of dark, linear necrosis starting from the root tip, resulting in the appearance of a characteristic “rat tail” (Figure 3). The inner bark and cambium necrosis progressively migrated towards the base of the stem, causing the bark to crack. Other nonspecific canopy symptoms, such as chlorosis, stunted growth, flower abortion and sudden death were also visible on these plants (Figure 1).



Figure 4. Symptoms on avocado nursery seedlings: brown apical and marginal leaf necrosis caused by *Colletotrichum* and *Neofusicoccum* spp. (a), black leaf lesions caused by *Phytophthora palmivora* (b), sunken canker with pycnidia on the stem lesion surface (c), *Phytophthora* dark bark necrosis (d), and *Phytophthora* root and collar rot (e–h).

In nurseries, the main symptoms observed on seedlings were brown apical and marginal leaf necrosis, sunken canker and dark necrosis on the stem, and root and collar rot (Figure 4).

3.2. Etiology

The symptomatic samples processed yielded in pure culture a total of 22 species, of which nine belonged to the *Botryosphaeriaceae* family (*Botryosphaeria dothidea*, *Diplodia mutila*, *D. seriata*, *Do. viticola*, *Macrophomina phaseolina*, *N. australe*, *N. cryptoaustrale*, *N. luteum* and *N. parvum*), 11 to the *Peronosporaceae* family (*P. bilorbang*, *P. cinnamomi*, *P. citricola*, *P. citrophthora*, *P. gonapodyides*, *P. multivora*, *P. nicotianae*, *P. niederhauserii*, *P. palmivora*, *P. polonica*, *P. plurivora*) and two to the *Glomerellaceae* family (*Colletotrichum gloeosporioides* and *C. siamense*) (Tables 3 and 4). On all plant organs, a variable and complex assemblage of pathogens was detected. On leaves showing large necrotic lesions, six species were isolated: in particular, *N. australe* from leaves of mature avocado trees and *P. palmivora* from leaves of nursery seedlings (Tables 3 and 4). From leaves with anthracnose symptoms collected in Sardinian orchards, the species *C. gloeosporioides* was consistently isolated, whereas the sister species *C. siamense* was isolated from leaves of seedlings with apical necrotic lesions in Sicily (Table 4).

Table 3. Number of isolates of each species obtained from orchard samples at each site.

Plant Organ	Symptom	Number of Isolates of Each Pathogenic Species	Site
Leaf	Necrosis	<i>Colletotrichum gloeosporioides</i> (12); <i>Diplodia seriata</i> (2), <i>Neofusicoccum australe</i> (20), <i>Neofusicoccum parvum</i> (7)	4, 5, 6
Fruit	Early dark rot	<i>N. australe</i> (9), <i>Phytophthora multivora</i> (1), <i>Phytophthora palmivora</i> (41)	4
	Stem-end rot	<i>N. australe</i> (3), <i>Neofusicoccum luteum</i> (3)	4
	Anthracnose	<i>Colletotrichum gloeosporioides</i> (2)	4
Shoot, branch and stem	Dark bark necrosis	<i>Phytophthora citrophthora</i> (6), <i>P. palmivora</i> (2)	4, 5
	Cankers with V-shaped sector	<i>Botryosphaeria dothidea</i> (2), <i>C. gloeosporioides</i> (9), <i>D. seriata</i> (12), <i>Dothiorella viticola</i> (3), <i>Macrophomina phaseolina</i> (4), <i>N. australe</i> (64), <i>Neofusicoccum cryptoaustrale</i> (3), <i>N. luteum</i> (11), <i>N. parvum</i> (14)	4, 5, 6
	Collar rot	<i>P. cinnamomi</i> (3), <i>P. citrophthora</i> (2)	4, 5
Root	Root rot	<i>Phytophthora bilorbang</i> (4), <i>Phytophthora cinnamomi</i> (22), <i>P. citrophthora</i> (11), <i>Phytophthora gonapodyides</i> (2), <i>P. multivora</i> (2), <i>Phytophthora nicotianae</i> (1), <i>Phytophthora niederhauserii</i> (5), <i>P. palmivora</i> (1), <i>Phytophthora plurivora</i> (3)	4, 5, 6

Table 4. Number of isolates of each species obtained from nursery samples at each site.

Plant organ	Symptom	Number of Isolates of Each Pathogenic Species	Site
Leaf	Necrosis	<i>Colletotrichum siamense</i> (4), <i>N. parvum</i> (2), <i>P. palmivora</i> (8)	1, 2
Shoot, branch and stem	Dark bark necrosis	<i>P. citrophthora</i> (2), <i>P. palmivora</i> (1)	1, 2
	Sunken cankers	<i>C. siamense</i> (3), <i>Diplodia mutila</i> (2), <i>N. parvum</i> (5), <i>Neofusicoccum luteum</i> (2)	1, 2
Root	Root rot	<i>P. cinnamomi</i> (2), <i>Phytophthora citricola</i> (2), <i>P. citrophthora</i> (4), <i>P. niederhauserii</i> (1), <i>P. palmivora</i> (3), <i>Phytophthora polonica</i> (1)	1, 2, 3

In the fruit, *P. palmivora* was the dominant species associated with early dark rot during the summer, whereas *N. australe* and *N. luteum* were from the stem-end rot symptoms in autumn (Table 3). Isolation from ripening avocado fruits with anthracnose symptoms yielded colonies of *C. gloeosporioides*.

Neofusicoccum australe and *N. parvum* were consistently associated with branch and stem cankers showing the V-shaped necrotic sector in a cross section (Table 3). The sunken appearance of *Botryosphaeriaceae* cankers with the V-shaped necrotic sector was more visible on twigs and branches, whereas on the main stem this was less clear due to an abundant production of dark-colored exudates by the plant. Sometimes a white exudate was also present in the center of the canker. In addition, on both branches and the main stem, outer and inner bark dark necrosis with blackish exudates without a sunken appearance were observed. From this type of bark canker from both nursery seedlings and mature trees, *P. citrophthora* and *P. palmivora* were consistently isolated.

From collar and root rot symptoms, a total of 11 *Phytophthora* species were isolated: nine from orchard samples and six from samples taken from nursery seedlings (Tables 3 and 4). Four species (*P. cinnamomi*, *P. citrophthora*, *P. niederhauserii* and *P. palmivora*) were present in both the nursery and orchard samples.

3.3. Phylogeny

Phylogenetic relationships among the representative *Colletotrichum*, *Phytophthora* and botryosphaeriaceous isolates obtained in this study and the closely formally described species were clarified using a single-locus analysis for *Phytophthora* spp. and a Multilocus Sequence Analysis (MLSA) based on concatenated sequences of ITS and *gapdh* regions for *Colletotrichum* species, and ITS plus *tef1-α* region for botryosphaeriaceous isolates.

Single and concatenated phylogenies are shown in Figures 5–7. The ML evolutionary reconstruction placed the representative *Phytophthora* isolates into 11 well-supported terminal clades (ML bootstrap, 100%) together with the relative ex-type culture (Figure 5). The 11 species were distributed into 6 of the 13 major *Phytophthora* clades [35].

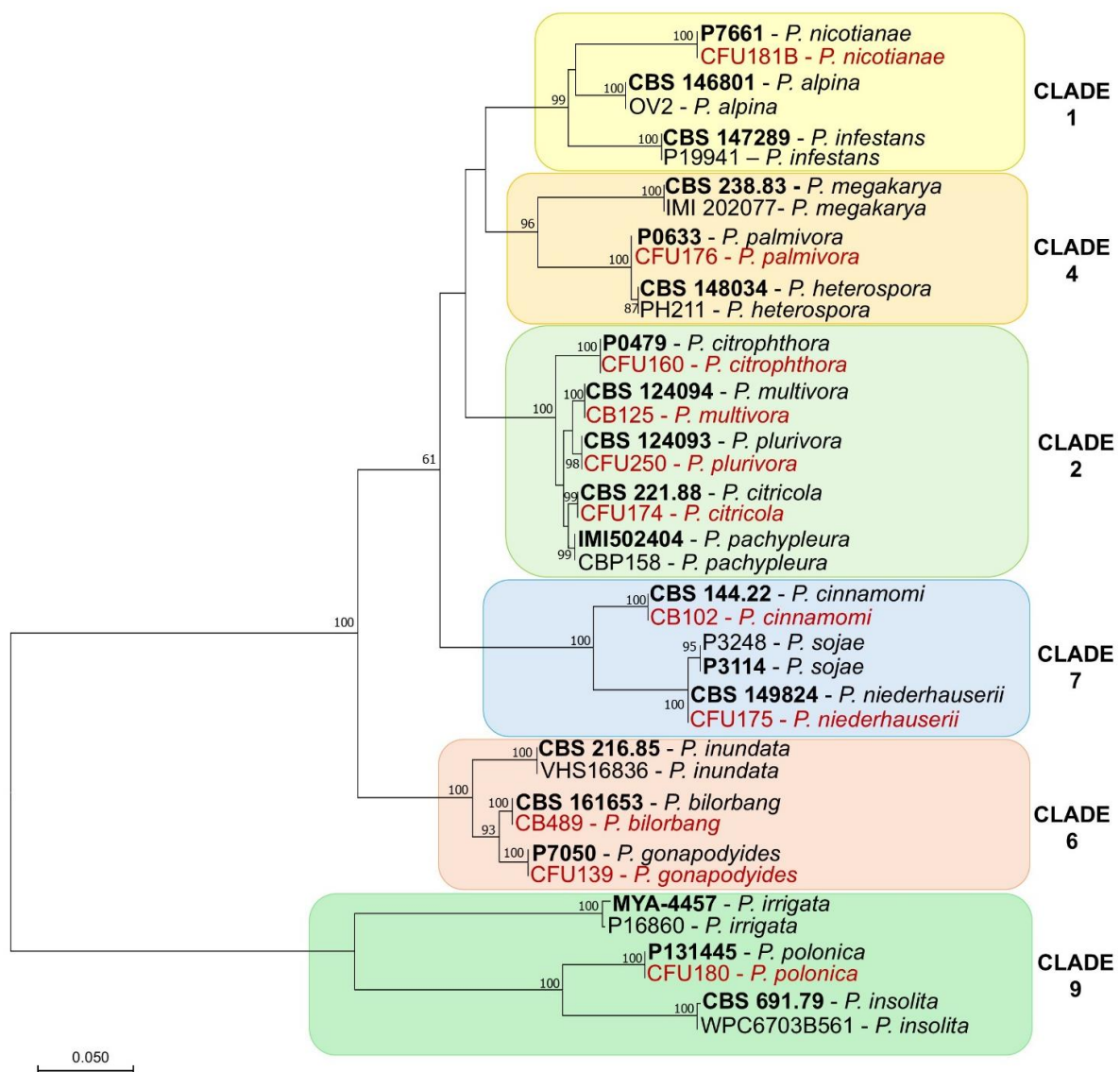


Figure 5. Maximum-likelihood tree based on ITS sequences of 40 *Phytophthora* isolates belonging to 20 species. Data are based on the General Time Reversible Model. A discrete Gamma distribution was used to model evolutionary rate differences among sites. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Bootstrap support values in percentages (1000 replicates) are given at the nodes. Ex-type cultures are in bold and representative isolates obtained in this study are in red.

ML analyses clustered the two representative *Colletotrichum* isolates into two terminal clades together with the ex-type culture (ICMP18578) of *C. siamense* and the ex-type culture (CBS 112999) of *C. gloeosporioides*, respectively (Figure 6), confirming the identification.

The representative botryosphaeriaceous isolates studied were distributed into nine terminal clades within five genera belonging to the *Botryosphaeriaceae* family (Figure 7). All terminal clades showed a high bootstrap support (ML, 91/100%) and included the ex-type culture of each species (Figure 7).

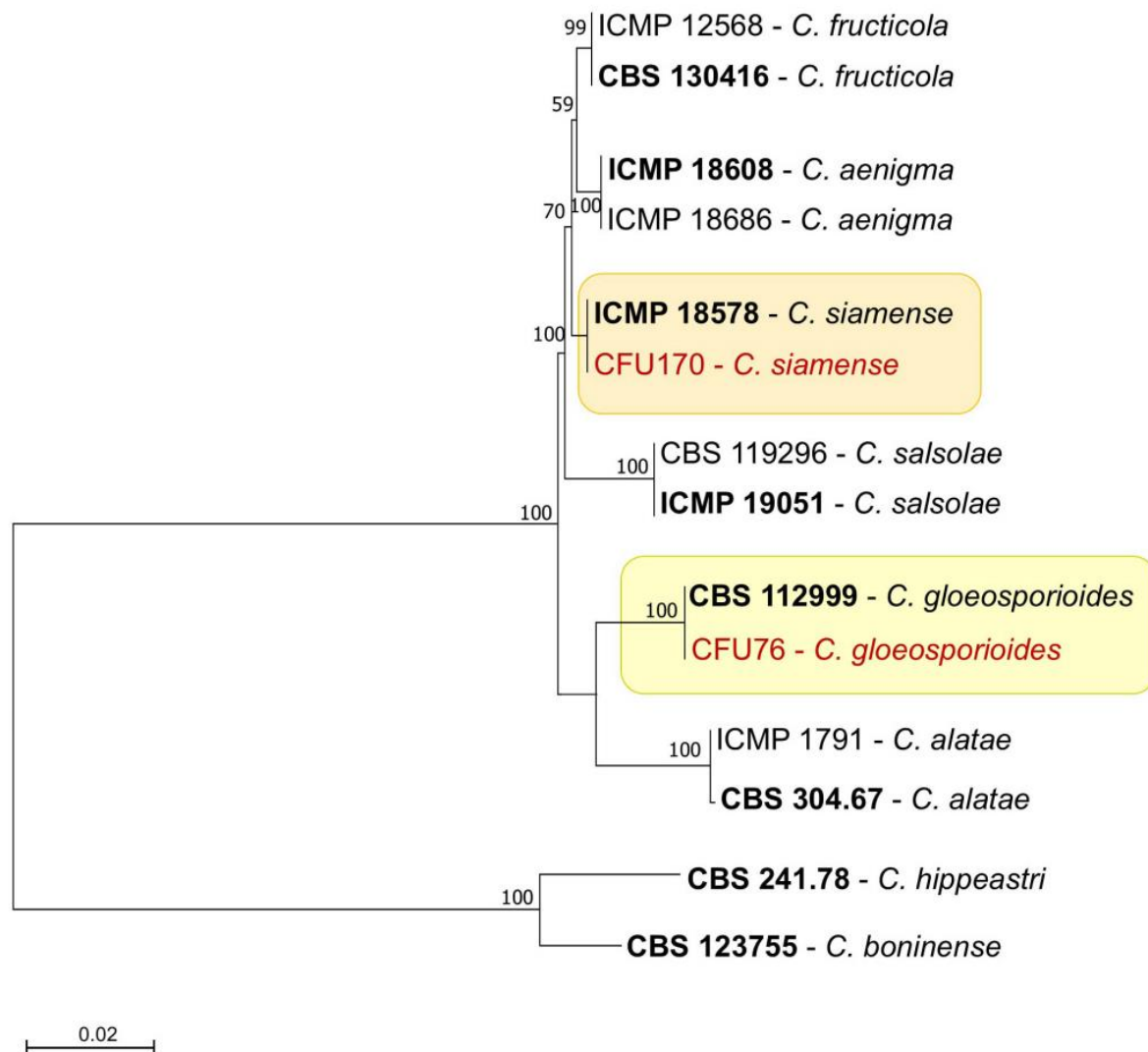


Figure 6. Maximum-likelihood tree obtained from concatenated ITS and *gapdh* sequences of 14 *Colletotrichum* isolates belonging to 8 species. Data are based on the General Time Reversible Model. A discrete Gamma distribution was used to model evolutionary rate differences among sites. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Bootstrap support values in percentages (1000 replicates) are given at the nodes. Ex-type cultures are in bold, whereas the representative isolates obtained in this study are in red.

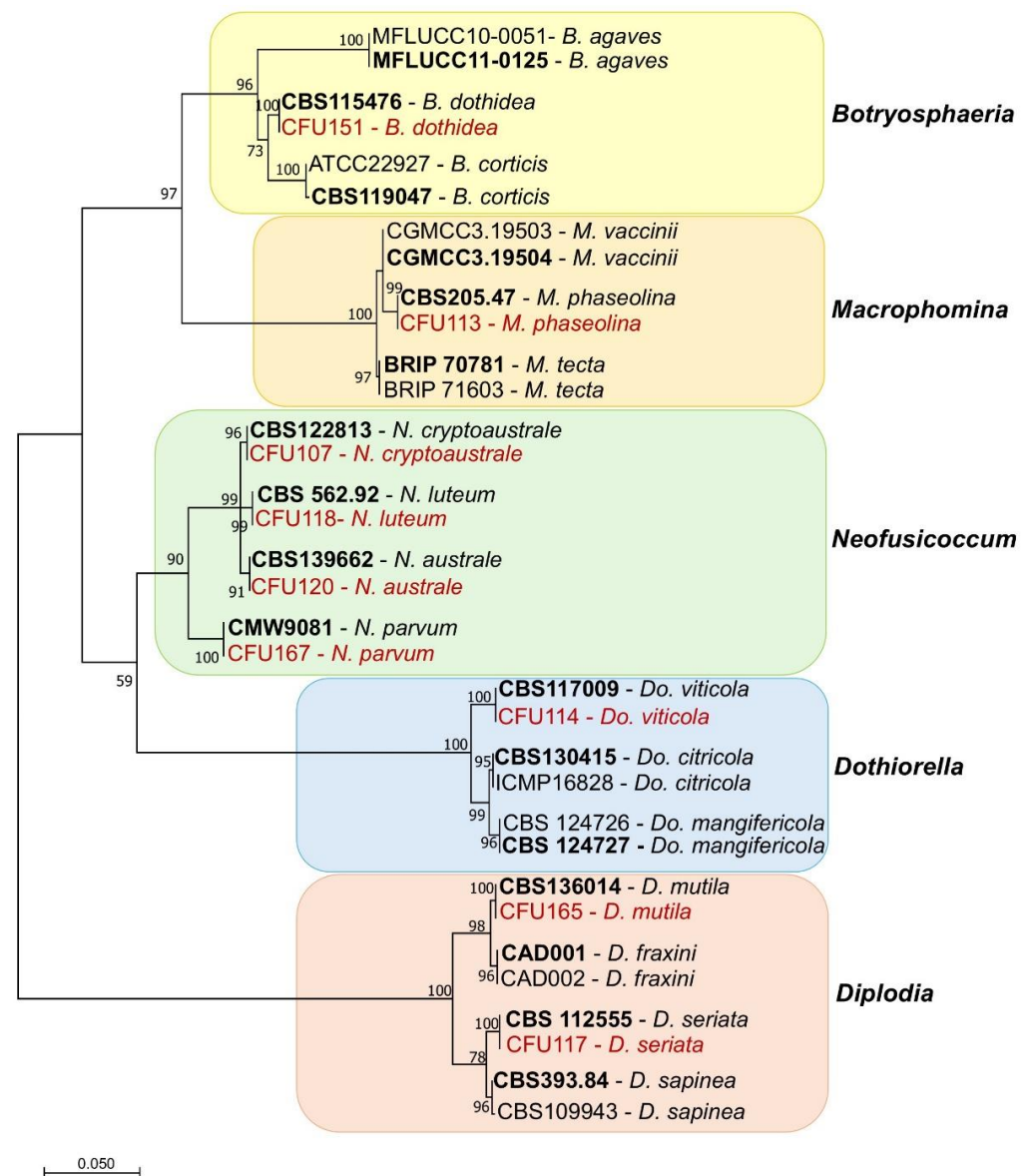


Figure 7. Maximum-likelihood tree obtained from concatenated ITS and *tef1-α* sequences of 34 botryosphaeriaceous isolates belonging to 17 species. Data are based on the General Time Reversible Model. A discrete Gamma distribution was used to model evolutionary rate differences among sites. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Bootstrap support values in percentage (1000 replicates) are given at the nodes. Ex-type cultures are in bold, whereas representative isolates obtained in this study are in red.

3.4. Pathogenicity

At the end of the experimental period of the first bioassay, all fungal and fungal-like isolates caused on stem external and internal necrotic lesions on the stem were consistent with those observed in nursery/orchard, whereas the control plants were asymptomatic (Figure 8). In particular, *C. siamense* produced the longest internal lesions, followed by *N. parvum*, *N. australe* and *P. cinnamomi*. Furthermore, lesions caused by *C. siamense* were significantly different from those of *Phytophthora* species, except for *P. cinnamomi*, while they did not differ significantly from those of *Botryosphaeriaceae*, except for *D. mutila* (Table 5).

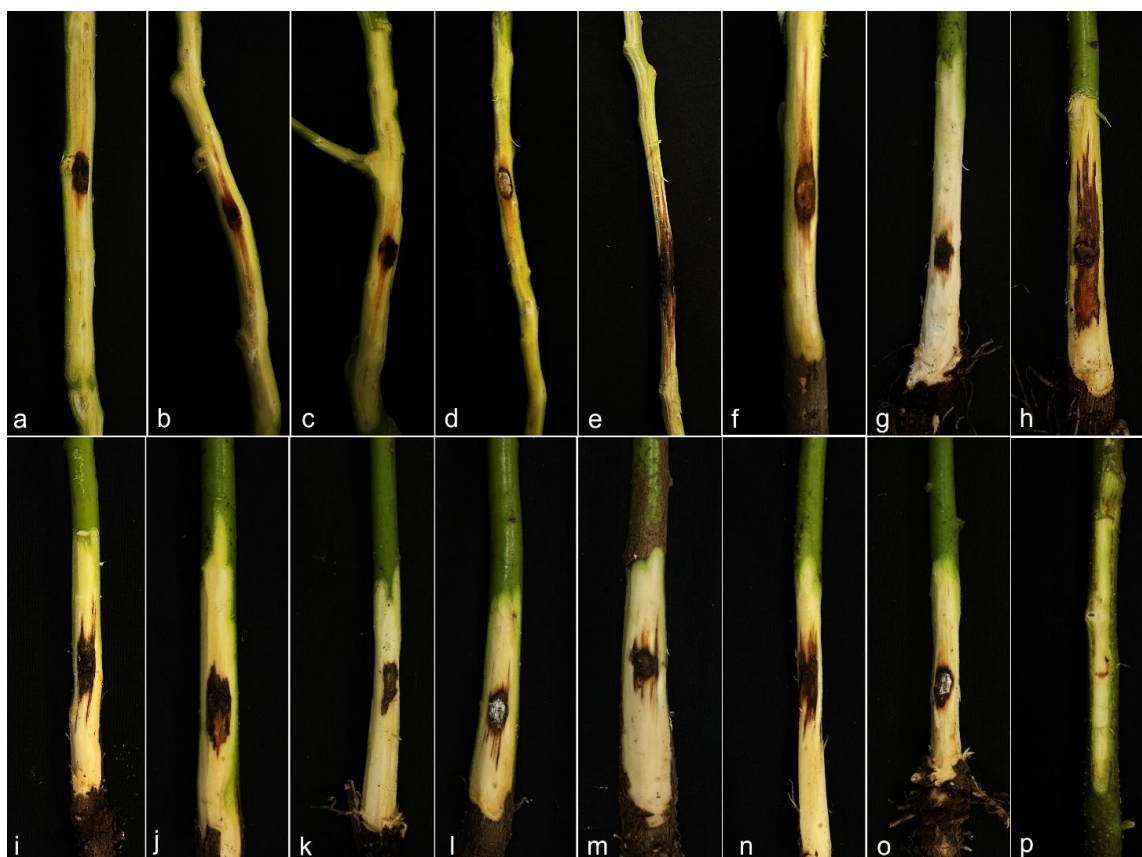


Figure 8. Result of the pathogenicity test on plant stem inoculated with *Diplodia mutila* (a), *Colletotrichum siamense* (b), *Diplodia seriata* (c), *Dothiorella viticola* (d), *Neofusicoccum parvum* (e), *Neofusicoccum australe* (f), *Phytophthora bilorbang* (g), *Phytophthora cinnamomi* (h), *Phytophthora citricola* (i), *Phytophthora citrophthora* (j), *Phytophthora gonapodyides* (k), *Phytophthora multivora* (l), *Phytophthora niederhauserii* (m), *Phytophthora palmivora* (n), and *Phytophthora polonica* (o) after 90 days. Control (p).

Table 5. Mean lesion size ($\pm$ standard deviation) caused by the different pathogens on avocado stems, leaves and fruits.

Pathogen	Necrosis Length on Stem (mm)	Average Lesion Size on Leaves (mm ²)	Necrosis Length on Fruit Exocarp (mm)	Necrosis Length on Fruit Mesocarp (mm)
<i>Colletotrichum siamense</i>	70 $\pm$ 39 a	6 $\pm$ 2 bc	nt	nt
<i>Diplodia mutila</i>	25 $\pm$ 20 bc	10 $\pm$ 4 b	nt	nt
<i>Diplodia seriata</i>	32 $\pm$ 34 abc	1 $\pm$ 0.2 bcd	nt	nt
<i>Dothiorella viticola</i>	30 $\pm$ 36 abc	5 $\pm$ 1 bcd	nt	nt
<i>Neofusicoccum australe</i>	55 $\pm$ 32 abc	27 $\pm$ 12 a	27 $\pm$ 10 b	61 $\pm$ 13 a
<i>Neofusicoccum parvum</i>	61 $\pm$ 36 ab	7 $\pm$ 1 bc	nt	nt
<i>Phytophthora bilorbang</i>	21 $\pm$ 8 bc	0 $\pm$ 0 d	nt	nt
<i>Phytophthora cinnamomi</i>	45 $\pm$ 10 abc	0 $\pm$ 0 d	nt	nt
<i>Phytophthora citricola</i>	22 $\pm$ 7 bc	0 $\pm$ 0 d	nt	nt
<i>Phytophthora citrophthora</i>	21 $\pm$ 4 bc	2 $\pm$ 1 cd	nt	nt
<i>Phytophthora gonapodyides</i>	14 $\pm$ 3 c	0 $\pm$ 0 d	nt	nt
<i>Phytophthora multivora</i>	20 $\pm$ 8 bc	2 $\pm$ 0.2 cd	nt	nt
<i>Phytophthora niederhauserii</i>	21 $\pm$ 4 bc	2 $\pm$ 2 cd	nt	nt
<i>Phytophthora palmivora</i>	24 $\pm$ 8 bc	8 $\pm$ 5 bc	70 $\pm$ 48 a	49 $\pm$ 10 a
<i>Phytophthora polonica</i>	17 $\pm$ 7 bc	0 $\pm$ 0 d	nt	nt
Control	-	-	-	-

Values in column with the same letter do not differ significantly at $p = 0.05$, according to LSD multiple range test. nt = not tested.

Among the *Phytophthora* species tested on avocado leaves, *P. palmivora* proved to be an aggressive leaf pathogen causing extensive necrotic lesions that were significantly different from those of the other *Phytophthora* species and control (Figure 9 and Table 5). *Phytophthora citrophthora*, *P. multivora* and *P. niederhauserii* caused lesions ranging from 1.53 mm to 1.67 mm. In contrast, *P. citricola*, *P. polonica*, *P. bilorbang*, and *P. gonapodyides* did not produce any leaf symptoms, while *P. cinnamomi* induced only small, dark lesions restricted to the inoculation sites (Table 5). Among the fungal species, all isolates caused necrotic lesions with *N. australe* recognized as the main aggressive species (Table 5).

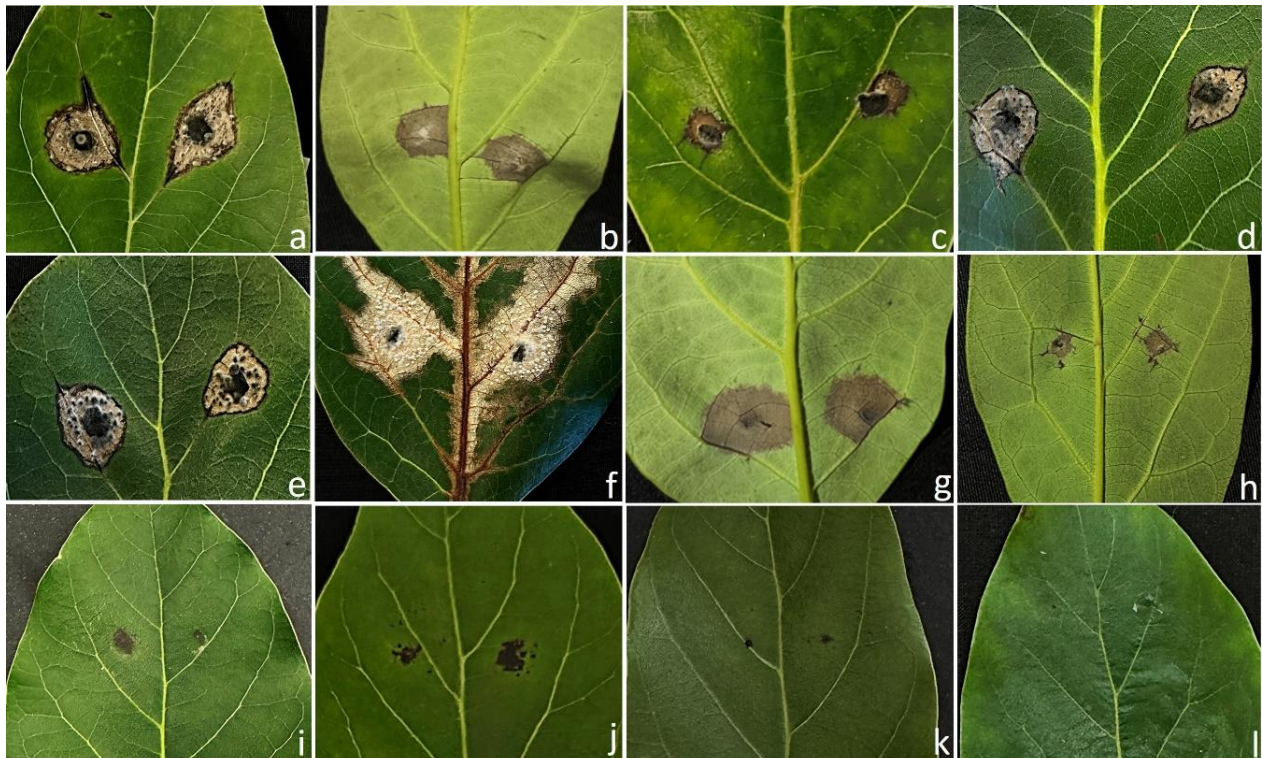


Figure 9. Result of the pathogenicity test on leaf inoculated with *Diplodia mutila* (a), *Colletotrichum siamense* (b), *Diplodia seriata* (c), *Dothiorella viticola* (d), *Neofusicoccum parvum* (e), *Neofusicoccum australe* (f), *Phytophthora palmivora* (g), *Phytophthora citrophthora* (h), *Phytophthora multivora* (i), *Phytophthora niederhauserii* (j), and *Phytophthora cinnamomi* (k) after 7 days. Control (l).

Phytophthora palmivora and *N. australe* proved to be pathogenic on avocado fruits, although with different aggressiveness. Statistically significant differences emerged among the two species in terms of lesion size and ability to colonize fruit tissues (Figure 10 and Table 5). In particular, *N. australe* caused extensive necrotic lesions that quickly expanded towards internal tissues (mesocarp), whereas *P. palmivora* caused chiefly superficial (exocarp) soft rot lesions that were easily recognizable by touch. Control fruits inoculated with sterile PDA plugs remained asymptomatic (Figure 10).

All species used in the pathogenicity bioassays were successfully re-isolated from the margin of necrotic lesions in all bioassays, thus fulfilling Koch's postulates.



Figure 10. Symptoms observed in avocado fruits (var. Bacon) inoculated with: *Neofusicoccum australe* (a,d) and *P. palmivora* (b,e) after four days. Control (c,f).

4. Discussion

In this study, we present a comprehensive overview of the pathogens involved in the etiology of emerging diseases affecting avocado in Italy. To address this major issue, we employed a combination of different diagnostic analyses on all plant organs, using various isolation techniques which allowed us to isolate and characterize 22 pathogenic species belonging to seven genera in the *Botryosphaeriaceae*, *Glomerellaceae* and *Peronosporaceae* families.

These results highlight that, in addition to several *Botryosphaeriaceae* and *Colletotrichum* species already known to be avocado pathogens in Italy, different species of *Phytophthora* are also involved in the etiology of the emerging avocado diseases, including some never before reported on avocado worldwide. The pathogenicity tests confirmed the aggressiveness of the different fungal and oomycetes species and provided important information about plant organs' susceptibility.

Among the *Botryosphaeriaceae* species, *N. parvum* was confirmed to be one of the main pathogens directly associated with leaf, branch and stem avocado disease symptoms in Italy [6]. *Neofusicoccum parvum* has also been reported on avocado in Turkey [36], Spain [37,38], Mexico [39], Chile [40] and California [41]. Over the past few years, an increasing number of studies have focused on the primary role played by *N. parvum* in emerging diseases on both agriculture [16,26,42,43] and forestry ecosystems [16,44–46] in Europe. In addition to *N. parvum*, two other *Botryosphaeriaceae* species, namely *N. australe* and *N. luteum*, were isolated with a high frequency from the symptomatic avocado samples. In particular, *N. australe*, reported here for the first time on avocado in Italy, was the dominant species on stem cankers. The high isolation frequency of *N. australe* obtained in

this study agrees with previous studies conducted in avocado orchards in California, Chile, Turkey, and Spain [38,47–49]. Genetic evidence strongly suggests that *N. australe* is native to Western Australia, where it exists as a common endophyte or opportunistic pathogen of eucalypts [50,51]. Given the widespread use of eucalyptus as a windbreak in Sardinian avocado orchards, a continuous host-jump of this pathogen between avocado and eucalypt is plausible.

The other *Botryosphaeriaceae* species (*B. dothidea*, *D. seriata*, *D. mutila*, *Do. viticola*, *M. phaseolina* and *N. cryptoaustrale*) were isolated with a lower frequency and especially from sunken cankers. Overall, our results reveal contrasting infection patterns among *Botryosphaeriaceae* species: while *N. australe*, *N. luteum* and *N. parvum* exhibit the ability to infect both woody tissues (branches and trunk) and herbaceous tissues (leaves, shoots, fruits), the other species display a specialization towards woody tissues. Among these species, *D. seriata*, *D. mutila*, and *Do. viticola* represent a first report on avocado in Italy.

Concerns like those regarding the impact of *Botryosphaeriaceae* have also emerged for the involvement of several *Phytophthora* species as avocado fruit rot, cankers, and root rot agents. These pathogens have also been associated for the first time with leaf diseases. Overall, nine different *Phytophthora* species were isolated and identified. Multiple species were often detected from a single sample and from the same site. The most consistently isolated species were *P. cinnamomi* from the roots and stem and *P. palmivora* from the fruits and leaves. Both species were isolated from mature trees on orchards and nursery plants, confirming the wide diffusion of these pathogens in Italian nurseries [52,53]. The international trade of plant material is widely recognized as the main pathway in the accidental introduction of *Phytophthora* species into agriculture, natural and urban ecosystems worldwide [54]. Once established in a new site (natural or anthropized), the impact of *Phytophthora* species is unpredictable and difficult to contrast [55,56]. Over the last 30 years, our knowledge about global *Phytophthora* diversity has improved with the number of culturable species exceeding 200 [57]. This result derives from the refinement of isolation and identification techniques, as well as from the constant discovery of new diseases caused by these pathogens [57]. The co-occurrence of different *Phytophthora* species associated with avocado root rot detected in this study could have a direct impact on avocado management strategies and in the screening for resistant rootstocks that are currently evaluated chiefly for *P. cinnamomi* [58,59]. Therefore, it is of the utmost importance to take this complex etiology into due consideration, since the *Phytophthora* species isolated from avocado are known to have optimal peaks of activity at different times of the year, as well as different infection and reproductive strategies [55].

Phytophthora cinnamomi is considered a highly invasive pathogen due to its virulence, wide host range, and ability to survive in the soil for long periods through the production of chlamydospores [56,60]. Avocado root rot caused by *P. cinnamomi* was first reported in Italy in 1998 on eight-year-old trees in Sicily [61]. Symptomatic trees were grafted on two different rootstocks including the G6, a Mexican selection reported to have some field resistance towards *P. cinnamomi* infections [58]. After this first report, there have been no other reports of *Phytophthora* affecting avocado in Italy before this work; therefore, the other species (*P. bilorbang*, *P. citricola*, *P. citrophthora*, *P. gonapodyides*, *P. multivora*, *P. nicotianae*, *P. niederhauserii*, *P. palmivora*, *P. polonica*, *P. plurivora*) are reported here for the first time on avocado in Italy. The involvement of *P. palmivora* in the fruit rot etiology is of particular concern. *Phytophthora palmivora* is a major pathogenic oomycete causing devastating fruit rot, stem canker and bud rot across different countries and tropical hosts such as cacao, coconut, durian, papaya and pomegranate [43,62–65]. Its ability to produce persistent and caducous sporangia support an aerial or soilborne lifestyle. In avocado, it causes a serious,

often rapid, fruit rot, often leaving a white mycelium on the fruit surface. Similar symptoms have recently been reported in Italy on pomegranate fruits [43].

Among the other *Phytophthora* species, four within the major clade 2 (*P. citricola*, *P. citrophthora*, *P. multivora* and *P. plurivora*) are highly polyphagous oomycetes with a broad host range. These species are notorious for causing serious root rot, collar rot and stem canker diseases on a variety of woody agricultural, ornamental and forest tree species such as alders, common ash, European beech, olive, paulownia and walnut trees in Italy [26,46,53,66–71]. Interestingly, *P. citrophthora* caused bleeding cankers on the trunk that were externally identical to those caused by *N. parvum*. Only through an invasive investigation involving a cross section of the trunk and observation of the internal wood symptoms was it possible to differentiate the attacks of the two pathogens. *Phytophthora citrophthora* is an aggressive pathogen involved in severe citrus diseases worldwide, including root rot, bleeding stem cankers (gummosis) and brown fruit rot [72]. In Italy, avocado cultivation is gradually expanding into areas where citrus fruit was grown until a few years ago [4]. This could pose a serious risk to the avocado industry, given the widespread distribution of *P. citrophthora* in Italian citrus groves [73].

To better understand the role of each *Phytophthora* species in the colonization of the avocado root system as well as on the potential synergistic interaction among these different species, an investigation based on the artificial inoculation of avocado seedlings with an infected soil mixed with the nine *Phytophthora* species is in progress.

Besides *Botryosphaeriaceae* and *Phytophthora* species, isolation from leaves, fruits and branch samples yielded fungal isolates belonging to the species *C. gloeosporioides* and *C. siamense*. *Colletotrichum* species are well-known pathogens that affect various agricultural crops in tropical, subtropical and temperate regions. Diseases caused by these pathogens can cause significant economic losses, especially in the postharvest phase of tropical fruits, including avocados. Several species within the *C. gloeosporioides* complex, including *C. gloeosporioides* s.s. and *C. siamense*, have already been reported on avocado in different countries [7,31]. In Italy, *C. gloeosporioides* s.s., *C. fructicola*, *C. fiorinae* and *C. perseae* have previously been reported as avocado pathogens, whereas this is the first report of *C. siamense* as an avocado pathogen [15]. *Colletotrichum siamense* was originally described as a saprophyte, endophyte and pathogen of coffee (*Coffea arabica*) in Thailand [74], and then reported as causing severe diseases on a wide range of hosts globally [75–79]. A pathogenicity test demonstrated that *C. siamense* is an aggressive avocado pathogen that is able to infect stem and leaf tissues. The occurrence of *Colletotrichum siamense*-related diseases in nursery seedlings raises concerns about the potential diffusion of this pathogen through propagation materials. Both *Colletotrichum* species were recovered from lesions on twigs and branches in association with *Neofusicoccum* species, suggesting a secondary role played by these species on avocado wood tissues. This agrees with what was previously reported by various studies in other countries [80,81].

5. Conclusions

Although avocado is emerging as a profitable crop in Italy, the co-occurrence of severe attacks by several *Botryosphaeriaceae* and *Phytophthora* species outlines an alarming scenario. The high pathogen diversity detected in avocado orchards appears to be the result of multiple introductions by infected nursery material and host jumps between adjacent fruit and forest trees and avocado.

Overall, *N. parvum* confirmed its wide distribution on avocado orchards in the Mediterranean area, whereas *P. australe* was recognized as the dominant species in Sardinia. *Phytophthora cinnamomi* is widely regarded as the most destructive avocado pathogen globally, but the results of this study highlight that several other species and in particular *P. palmivora*

and *P. citrophthora* represent an emerging threat to avocado in Italy. The different species of *Phytophthora* can infect all aerial organs, albeit with different specificity, often causing symptoms that are apparently identical when observed externally to those of *Botryosphaeriaceae* on the trunk. This complex etiology and the simultaneous involvement of several ecologically unrelated pathogens represent a major challenge for the development of adequate management strategies. At the same time, this issue raises the question about the limit of a diagnosis priority based on the new technologies offered by artificial intelligence (AI) and machine learning (ML) that analyzes plant images to automatically diagnose plant diseases in real time.

Author Contributions: Conceptualization, B.T.L., C.B. and D.A.; methodology, B.T.L., C.B. and D.A.; validation, B.T.L., C.B. and D.A.; formal analysis, B.T.L., C.B., J.M., L.M., D.A., L.V. and G.P.; investigation, B.T.L., C.B., J.M., L.M., D.A., L.V. and G.P.; resources, B.T.L., C.B. and D.A.; data curation, B.T.L., C.B., L.M., D.A. and L.V.; writing—original draft preparation, B.T.L.; writing—review and editing, B.T.L., C.B., L.M., D.A., L.V. and G.P.; supervision, B.T.L., C.B., L.M. and D.A.; funding acquisition, B.T.L., C.B. and D.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Associazione Agricoltori Villacidresi, and partially by the Project BIRD 2025 TESAF1SIDPROGETTI-00027 (Dept. TESAF, University of Padua, Italy), by the University of Padova with the projects: 2025TESAF1DOR-00255 and 2025TESAF1DOR-00262, by Piano di incentivi per la ricerca di Ateneo DIME-SIECO 2024-2026, University of Catania (Italy) and from the University of Catania (PhD grant to Laura Vecchio).

Data Availability Statement: The original contributions presented in this study are included in the article. Further requests can be directed to the corresponding author.

Acknowledgments: The authors thank Giorgio Ionta and Giovanni Ignazio Muscas for their logistic support and technical assistance during field surveys and sampling.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Silva, T.R.; Ledesma, N. Avocado history, biodiversity and production. In *Sustainable Horticultural Systems: Issues, Technology and Innovation*; Nandwani, D., Ed.; Springer International Publishing: Cham, Switzerland, 2014; pp. 157–205. [\[CrossRef\]](#)
2. Cervantes-Paz, B.; Yahia, E.M. Avocado oil: Production and market demand, bioactive components, implications in health, and tendencies and potential uses. *Compr. Rev. Food Sci. Food Saf.* **2021**, *20*, 4120–4158. [\[CrossRef\]](#)
3. Huang, K.M.; Guan, Z.; Blare, T.; Hammami, A.M. Global avocado boom. *Choices* **2023**, *38*, 1–9.
4. Coldiretti. Italia tropicale, avocado, mango e guava avanzano nelle campagne. *Il Punto Coldiretti* **2025**.
5. Fiorenza, A.; Gusella, G.; Vecchio, L.; Aiello, D.; Polizzi, G. Diversity of *Botryosphaeriaceae* species associated with canker and dieback of avocado (*Persea americana*) in Italy. *Phytopathol. Mediterr.* **2023**, *62*, 47–63. [\[CrossRef\]](#)
6. Guarnaccia, V.; Vitale, A.; Cirvilleri, G.; Aiello, D.; Susca, A.; Epifani, F.; Polizzi, G. Characterisation and pathogenicity of fungal species associated with branch cankers and stem-end rot of avocado in Italy. *Eur. J. Plant Pathol.* **2016**, *146*, 963–976. [\[CrossRef\]](#)
7. Sharma, G.; Maymon, M.; Freeman, S. Epidemiology, pathology and identification of *Colletotrichum*, including a novel species associated with avocado (*Persea americana*) anthracnose in Israel. *Sci. Rep.* **2017**, *7*, 15839. [\[CrossRef\]](#) [\[PubMed\]](#)
8. Möller, H.; Slippers, B.; van den Berg, N. Branch canker battles: Understanding and managing the *Botryosphaeriaceae* in avocado. *Phytoparasitica* **2025**, *53*, 17. [\[CrossRef\]](#)
9. Ramírez-Gil, J.G.; Castañeda-Sánchez, D.A.; Morales-Osorio, J.G. Production of avocado trees infected with *Phytophthora cinnamomi* under different management regimes. *Plant Pathol.* **2017**, *66*, 623–632. [\[CrossRef\]](#)
10. Zakaria, L. Diversity of *Colletotrichum* species associated with anthracnose disease in tropical fruit crops—A review. *Agriculture* **2021**, *11*, 297. [\[CrossRef\]](#)
11. Pérez-Jiménez, R.M. Significant avocado diseases caused by fungi and oomycetes. *Eur. J. Plant Sci. Biotechnol.* **2008**, *2*, 1–24.
12. Kimaru, K.S.; Muchemi, K.P.; Mwangi, J.W. Effects of anthracnose disease on avocado production in Kenya. *Cogent Food Agric.* **2020**, *6*, 1799531. [\[CrossRef\]](#)

13. Dissanayake, D.M.S.; Adikaram, N.K.B.; Yakandawala, D.M.D.; Jayasinghe, L. Molecular phylogeny-based identification of *Colletotrichum endophytica* and *C. siamense* as causal agents of avocado anthracnose in Sri Lanka. *Ceylon J. Sci.* **2021**, *50*, 449–458. [\[CrossRef\]](#)
14. Martino, I.; Sorrentino, R.; Piccirillo, G.; Battaglia, V.; Polizzi, G.; Guarnaccia, V.; Lahoz, E. *Colletotrichum fioriniae*, causal agent of postharvest avocado fruit rot in Southern Italy. *Phytopathol. Mediterr.* **2024**, *63*, 375–383. [\[CrossRef\]](#)
15. Vecchio, L.; Martino, I.; Guarnaccia, V.; Polizzi, G.; Aiello, D. *Colletotrichum perseae* and *Colletotrichum gloeosporioides* sensu strictu Causing Stem Lesion and Dieback in Avocado in Italy. *Horticolturae* **2026**, *12*, 111. [\[CrossRef\]](#)
16. Aiello, D.; Bregant, C.; Carlucci, A.; Guarnaccia, V.; Gusella, G.; Linaldeddu, B.T.; Polizzi, G. Current status of Botryosphaeriaceae species in Italy: Impacts on agricultural crops and forest ecosystems. *Phytopathol. Mediterr.* **2023**, *62*, 381–412. [\[CrossRef\]](#)
17. Belisle, R.J.; McKee, B.; Hao, W.; Crowley, M.; Arpaia, M.L.; Miles, T.D.; Adaskaveg, J.E.; Manosalva, P. Phenotypic characterization of genetically distinct *Phytophthora cinnamomi* isolates from avocado. *Phytopathology* **2019**, *109*, 384–394. [\[CrossRef\]](#)
18. Zentmyer, G.A. Avocado diseases. *Int. J. Pest Manag.* **1984**, *30*, 388–400. [\[CrossRef\]](#)
19. Kurbetli, İ.; Sülü, G.; Aydoğdu, M.; Woodward, S.; Bayram, S. Outbreak of *Phytophthora cinnamomi* causing severe decline of avocado trees in southern Turkey. *J. Phytopathol.* **2020**, *168*, 533–541. [\[CrossRef\]](#)
20. Hong, C.X.; Gallegly, M.E.; Browne, G.T.; Bhat, R.G.; Richardson, P.A.; Kong, P. The avocado subgroup of *Phytophthora citricola* constitutes a distinct species, *Phytophthora mendei* sp. nov. *Mycologia* **2009**, *101*, 833–840. [\[CrossRef\]](#)
21. Mondragón-Flores, A.; Soto-Plancarte, A.; Rodríguez-Alvarado, G.; Manosalva, P.; Ochoa-Ascencio, S.; Hoyt, B.; Fernández-Pavía, S.P. First report of *Phytophthora mendei* causing root rot and canker in avocado (*Persea americana*) in Michoacán, Mexico. *Microorganisms* **2025**, *13*, 1471. [\[CrossRef\]](#) [\[PubMed\]](#)
22. Palacios-Joya, L.P.; Rodríguez-Arévalo, K.A.; Martínez, M.F.; Murcia-Riaño, N.; Rodríguez-Mora, D.M. *Phytophthora* species causing root rot in avocado seedlings at Colombian nurseries: Morphological, molecular, and pathogenic analysis. *Sci. Agropecu.* **2025**, *16*, 113–121. [\[CrossRef\]](#)
23. Kankavee, P.; Withee, P.; Haituk, S.; Wongwan, T.; Promthep, T.; Karunarathna, A.; Cheewangkoon, R. First report of *Phytophthora nicotianae* causing avocado root rot in Thailand. *J. Gen. Plant Pathol.* **2024**, *90*, 51–58. [\[CrossRef\]](#)
24. Rodríguez-Padron, C.; Siverio, F.; Perez-Sierra, A.; Rodriguez, A. Isolation and pathogenicity of *Phytophthora* species and *Phytophthora vexans* recovered from avocado orchards in the Canary Islands, including *Phytophthora niederhauserii* as a new pathogen of avocado. *Phytopathol. Mediterr.* **2018**, *57*, 89–106.
25. Machado, M.; Collazo, C.; Peña, M.; Renaud, M.A.; López, M.O.; Coto, O.; Zamora, V.; Cabrera, R.I.; Aranguren, M.; Boland, G.J. First report of *Phytophthora palmivora* Butler causing root rot on avocado (*Persea americana* Mill.) in Cuba. *Can. J. Plant Pathol.* **2012**, *34*, 323–348.
26. Linaldeddu, B.T.; Rossetto, G.; Maddau, L.; Vatrano, T.; Bregant, C. Diversity and pathogenicity of Botryosphaeriaceae and *Phytophthora* species associated with emerging olive diseases in Italy. *Agriculture* **2023**, *13*, 1575. [\[CrossRef\]](#)
27. Linderman, R.G.; Zeitoun, F. *Phytophthora cinnamomi* causing root rot and wilt of nursery- grown native Western azalea and Salal. *Plant Dis. Report.* **1977**, *61*, 1045–1048.
28. White, T.J.; Bruns, T.; Lee, S.; Taylor, J. Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. In *PCR Protocols: A Guide to Methods and Applications*; Innis, M.A., Gelfand, D.H., Sninsky, J.J., White, T.J., Eds.; Academic Press: San Diego, CA, USA, 1990; pp. 315–322.
29. Inderbitzin, P.; Bostock, R.M.; Trouillas, F.P.; Michailides, T.J. A six-locus phylogeny reveals high species diversity in Botryosphaeriaceae from California almond. *Mycologia* **2010**, *102*, 1350–1368. [\[CrossRef\]](#)
30. Guerber, J.C.; Liu, B.; Correll, J.C.; Johnston, P.R. Characterization of diversity in *Colletotrichum acutatum* sensu lato by sequence analysis of two gene introns, mtDNA and intron RFLPs, and mating compatibility. *Mycologia* **2003**, *95*, 872–895. [\[CrossRef\]](#) [\[PubMed\]](#)
31. Weir, B.S.; Johnston, P.R.; Damm, U. The *Colletotrichum gloeosporioides* species complex. *Stud. Mycol.* **2012**, *73*, 115–180. [\[CrossRef\]](#)
32. Thompson, J.D.; Gibson, T.J.; Plewniak, F.; Jeanmougin, F.; Higgins, D.G. The CLUSTAL_X windows interface: Flexible strategies for multiple sequence alignment aided by quality analysis tools. *Nucleic Acids Res.* **1997**, *25*, 4876–4882. [\[CrossRef\]](#)
33. Hall, T.A. BioEdit: A user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. In *Nucleic Acids Symposium Series*; Oxford University Press: Oxford, UK, 1999; Volume 41, pp. 95–98.
34. Kumar, S.; Stecher, G.; Li, M.; Knyaz, C.; Tamura, K. MEGA X: Molecular evolutionary genetics analysis across computing platforms. *Mol. Biol. Evol.* **2018**, *35*, 1547–1549. [\[CrossRef\]](#)
35. Bourret, T.B.; Choudhury, R.A.; Mehl, H.K.; Blomquist, C.L.; McRoberts, N.; Rizzo, D.M. Multiple origins of downy mildews and mito-nuclear discordance within the paraphyletic genus *Phytophthora*. *PLoS ONE* **2018**, *13*, e0192502. [\[CrossRef\]](#)
36. Çalış, Ö.; Çelik, S.; Fidan, H.; Tek, M.I.; Shah, M.; Tozlu, I.; Wani, S.H. Emerging pathogens and disease dynamics threatening avocado production in southern Türkiye. *J. Plant Dis. Prot.* **2024**, *131*, 1653–1663. [\[CrossRef\]](#)
37. Zea-Bonilla, T.; González-Sánchez, M.A.; Martín-Sánchez, P.M.; Pérez-Jiménez, R.M. Avocado dieback caused by *Neofusicoccum parvum* in the Andalucía region, Spain. *Plant Dis.* **2007**, *91*, 1052. [\[CrossRef\]](#)

38. Arjona-Girona, I.; Ruano-Rosa, D.; López-Herrera, C.J. Identification, pathogenicity and distribution of the causal agents of dieback in avocado orchards in Spain. *Span. J. Agric. Res.* **2019**, *17*, e1003. [\[CrossRef\]](#)
39. Molina-Gayosso, E.; Silva-Rojas, H.V.; García-Morales, S.; Avila-Quezada, G. First report of black spots on avocado fruit caused by *Neofusicoccum parvum* in Mexico. *Plant Dis.* **2012**, *96*, 287. [\[CrossRef\]](#)
40. Valencia, A.L.; Gil, P.M.; Latorre, B.A.; Rosales, I.M. Characterization and pathogenicity of Botryosphaeriaceae species obtained from avocado trees with branch canker and dieback and from avocado fruit with stem end rot in Chile. *Plant Dis.* **2019**, *103*, 996–1005. [\[CrossRef\]](#)
41. McDonald, V.; Lynch, S.; Eskalen, A. First report of *Neofusicoccum australe*, *N. luteum*, and *N. parvum* associated with avocado branch canker in California. *Plant Dis.* **2009**, *93*, 967. [\[CrossRef\]](#)
42. Thomidis, T.; Michailides, T.J.; Exadaktylou, E. *Neofusicoccum parvum* associated with fruit rot and shoot blight of peaches in Greece. *Eur. J. Plant Pathol.* **2011**, *131*, 661–668. [\[CrossRef\]](#)
43. Bregant, C.; Rossetto, G.; Montecchio, L.; Tundo, S.; Raiola, A.; Linaldeddu, B.T. First report of *Neofusicoccum parvum* and *Phytophthora palmivora* causing fruit rot of pomegranate in Italy. *Ital. J. Mycol.* **2024**, *53*, 65–74. [\[CrossRef\]](#)
44. Seddaiu, S.; Mello, A.; Sechi, C.; Cerboneschi, A.; Linaldeddu, B.T. First report of *Neofusicoccum parvum* associated with chestnut nut rot in Italy. *Plant Dis.* **2021**, *105*, 3743. [\[CrossRef\]](#)
45. Bregant, C.; Rossetto, G.; Linaldeddu, B.T.; Batista, E.; Hilário, S.; Alves, A. First report of *Neofusicoccum parvum* causing canker and dieback on *Alnus glutinosa* in Portugal. *New Dis. Rep.* **2023**, *47*, e12169. [\[CrossRef\]](#)
46. Bregant, C.; Marcolongo, M.; Montecchio, L.; Ogris, N.; Piškur, B.; Linaldeddu, B.T. Diversity and pathogenicity of Botryosphaeriaceae and *Phytophthora* species involved in emerging diseases of *Fagus sylvatica* in Italy and Slovenia. *Ital. J. Mycol.* **2024**, *53*, 166–177. [\[CrossRef\]](#)
47. Montealegre, J.R.; Ramírez, M.; Riquelme, D.; Armengol, J.; León, M.; Pérez, L.M. First report of *Neofusicoccum australe* in Chile causing avocado stem-end rot. *Plant Dis.* **2016**, *100*, 1018. [\[CrossRef\]](#)
48. Auger, J.; Palma, F.; Pérez, I.; Esterio, M. First report of *Neofusicoccum australe* (*Botryosphaeria australis*) as a branch dieback pathogen of avocado trees in Chile. *Plant Dis.* **2013**, *97*, 842. [\[CrossRef\]](#)
49. Akgül, D.S.; Awan, Q.N.; Güler, P.G.; Önelge, N. First report of anthracnose and stem end rot diseases caused by *Colletotrichum gloeosporioides* and *Neofusicoccum australe* on avocado fruits in Turkey. *Plant Dis.* **2016**, *100*, 1792. [\[CrossRef\]](#)
50. Burgess, T.I.; Sakalidis, M.L.; Hardy, G.E.S. Gene flow of the canker pathogen *Botryosphaeria australis* between *Eucalyptus globulus* plantations and native eucalypt forests in Western Australia. *Austral Ecol.* **2006**, *31*, 559–566. [\[CrossRef\]](#)
51. Burgess, T.I.; Tan, Y.P.; Garnas, J.; Edwards, J.; Scarlett, K.A.; Shuttleworth, L.A.; Daniel, R.; Dann, E.K.; Parkinson, L.E.; Dinh, Q.; et al. Current status of the Botryosphaeriaceae in Australia. *Australas. Plant Pathol.* **2019**, *48*, 35–44. [\[CrossRef\]](#)
52. Bregant, C.; Mulas, A.A.; Rossetto, G.; Deidda, A.; Maddau, L.; Piras, G.; Linaldeddu, B.T. *Phytophthora mediterranea* sp. nov., a new species closely related to *Phytophthora cinnamomi* from nursery plants of *Myrtus communis* in Italy. *Forests* **2021**, *12*, 682. [\[CrossRef\]](#)
53. Antonelli, C.; Biscontri, M.; Tabet, D.; Vettraino, A.M. The never-ending presence of *Phytophthora* species in Italian nurseries. *Pathogens* **2022**, *12*, 15. [\[CrossRef\]](#) [\[PubMed\]](#)
54. Brasier, C.M. The biosecurity threat to the UK and global environment from international trade in plants. *Plant Pathol.* **2008**, *57*, 792–808. [\[CrossRef\]](#)
55. Hansen, E.M.; Reeser, P.W.; Sutton, W. *Phytophthora* beyond agriculture. *Annu. Rev. Phytopathol.* **2012**, *50*, 359–378. [\[CrossRef\]](#) [\[PubMed\]](#)
56. Burgess, T.I.; Scott, J.K.; McDougall, K.L.; Stukely, M.J.; Crane, C.; Dunstan, W.A.; Brigg, F.; Andjic, V.; White, D.; Rudman, T.; et al. Current and projected global distribution of *Phytophthora cinnamomi*, one of the world's worst plant pathogens. *Glob. Chang. Biol.* **2017**, *23*, 1661–1674. [\[CrossRef\]](#) [\[PubMed\]](#)
57. Sarker, S.R.; Burgess, T.I.; Hardy, G.E.S.J.; McComb, J. Closing the Gap between the number of *Phytophthora* species isolated through baiting a soil sample and the number revealed through metabarcoding. *Mycol. Prog.* **2023**, *22*, 39. [\[CrossRef\]](#)
58. Coffey, M.D. *Phytophthora* root rot of Avocado. *Plant Dis.* **1987**, *71*, 1046–1052.
59. Dolan, T.E.; Coffey, M.D. Laboratory screening technique for assessing resistance of four avocado rootstocks to *Phytophthora cinnamomi*. *Plant Dis.* **1986**, *70*, 115–118. [\[CrossRef\]](#)
60. Bregant, C.; Batista, E.; Hilário, S.; Linaldeddu, B.T.; Alves, A. Diversity and distribution of *Phytophthora* species along an elevation gradient in natural and semi-natural forest ecosystems in Portugal. *Pathogens* **2025**, *14*, 103. [\[CrossRef\]](#) [\[PubMed\]](#)
61. Cacciola, S.O.; Pane, A.; Davino, M.; Magnano di San Lio, G. First report of root rot caused by *Phytophthora cinnamomi* on avocado in Italy. *Plant Dis.* **1998**, *82*, 1281. [\[CrossRef\]](#)
62. Puig, A.S.; Quintanilla, W.; Matsumoto, T.; Keith, L.; Gutierrez, O.A.; Marelli, J.P. *Phytophthora palmivora* causing disease on *Theobroma cacao* in Hawaii. *Agriculture* **2021**, *11*, 396. [\[CrossRef\]](#)

63. Blaha, G.; Hall, G.; Warokka, J.S.; Concibido, E.; Ortiz-Garcia, C. *Phytophthora* isolates from coconut plantations in Indonesia and Ivory Coast: Characterization and identification by morphology and isozyme analysis. *Mycol. Res.* **1994**, *98*, 1379–1389. [\[CrossRef\]](#)
64. Vawdrey, L.L.; Langdon, P.; Martin, T. Incidence and pathogenicity of *Phytophthora palmivora* and *Pythium vexans* associated with durian decline in far northern Queensland. *Australas. Plant Pathol.* **2005**, *34*, 127–128. [\[CrossRef\]](#)
65. Jami, F.; Botha, W.J.; Bhiya, T.; Ntuli, I.; Phakalatsane, T.; Schoeman, M.H.; Daneel, M. *Phytophthora palmivora*: A serious threat to papaya production in South Africa. *Eur. J. Plant Pathol.* **2026**. [\[CrossRef\]](#)
66. Schoebel, C.N.; Stewart, J.; Gruenwald, N.J.; Rigling, D.; Prospero, S. Population history and pathways of spread of the plant pathogen *Phytophthora plurivora*. *PLoS ONE* **2014**, *9*, e85368. [\[CrossRef\]](#) [\[PubMed\]](#)
67. Tsykun, T.; Prospero, S.; Schoebel, C.N.; Rea, A.; Burgess, T.I. Global invasion history of the emerging plant pathogen *Phytophthora multivora*. *BMC Genom.* **2022**, *23*, 153. [\[CrossRef\]](#)
68. Bregant, C.; Sanna, G.P.; Bottos, A.; Maddau, L.; Montecchio, L.; Linaldeddu, B.T. Diversity and pathogenicity of *Phytophthora* species associated with declining alder trees in Italy and description of *Phytophthora alpina* sp. nov. *Forests* **2020**, *11*, 848. [\[CrossRef\]](#)
69. Carloni, F.; Bregant, C.; Linaldeddu, B.T.; Murolo, S. First report of *Phytophthora acerina* and *Phytophthora palmivora* causing root rot, bleeding cankers and dieback of English walnut in Italy. *New Dis. Rep.* **2024**, *49*, e12259. [\[CrossRef\]](#)
70. Benigno, A.; Bregant, C.; Aglietti, C.; Rossetto, G.; Tolio, B.; Moricca, S.; Linaldeddu, B.T. Pathogenic fungi and oomycetes causing dieback on *Fraxinus* species in the Mediterranean climate change hotspot region. *Front. For. Glob. Change* **2023**, *6*, 1253022. [\[CrossRef\]](#)
71. Bregant, C.; Carloni, F.; Balestra, M.; Linaldeddu, B.T.; Murolo, S. Pathogenicity of *Botryosphaeriaceae* and *Phytophthora* species associated with *Paulownia* dieback, canker and root rot in Italy. *Phytopathol. Mediterr.* **2023**, *62*, 481–488. [\[CrossRef\]](#)
72. Hussain, T.; Legrifi, I.; El Magur, S.; Abdellatif, E.; Besselma, N.; Ait Barka, E.; Lahlali, R. *Phytophthora* spp.'s effect on citrus industry: Current status, challenges, and emerging control strategies. *CABI Rev.* **2025**, *20*, 0020. [\[CrossRef\]](#)
73. Ippolito, A.; Schena, L.; Nigro, F.; Ligorio, V.; Yaseen, T. Real-time detection of *Phytophthora nicotianae* and *P. citrophthora* in citrus roots and soil. *Eur. J. Plant Pathol.* **2004**, *110*, 833–843. [\[CrossRef\]](#)
74. Prihastuti, H.; Cai, L.; Chen, H.; McKenzie, E.H.C.; Hyde, K.D. Characterization of *Colletotrichum* species associated with coffee berries in northern Thailand. *Fungal Divers.* **2009**, *39*, 89–109.
75. Jayawardena, R.S.; Hyde, K.D.; Damm, U.; Cai, L.; Liu, M.; Li, X.H.; Yan, J.Y. Notes on currently accepted species of *Colletotrichum*. *Mycosphere* **2016**, *7*, 1192–1260. [\[CrossRef\]](#)
76. Zhang, L.; Zhao, L.; Liang, C.; Yu, L.; Zhang, Y. *Colletotrichum* species (Glomerellales, Glomerellaceae) causing walnut anthracnose in China. *MycKeys* **2024**, *108*, 95–113. [\[CrossRef\]](#)
77. Cao, X.; Xu, X.; Che, H.; West, J.S.; Luo, D. Three *Colletotrichum* species, including a new species, are associated with leaf anthracnose of rubber tree in Hainan, China. *Plant Dis.* **2019**, *103*, 117–124. [\[CrossRef\]](#)
78. Cao, X.R.; Xu, X.M.; Che, H.Y.; West, J.S.; Luo, D.Q. Characteristics and distribution of *Colletotrichum* species in coffee plantations in Hainan, China. *Plant Pathol.* **2019**, *68*, 1146–1156. [\[CrossRef\]](#)
79. Cao, X.; Xu, X.; Che, H.; West, J.S.; Luo, D. Eight *Colletotrichum* species, including a novel species, are associated with areca palm anthracnose in Hainan, China. *Plant Dis.* **2020**, *104*, 1369–1377. [\[CrossRef\]](#) [\[PubMed\]](#)
80. Guarnaccia, V.; Kraus, C.; Markakis, E.; Alves, A.; Armengol, J.; Eichmeier, A.; Compant, S.; Gramaje, D. Fungal trunk diseases of fruit trees in Europe: Pathogens, spread and future directions. *Phytopathol. Mediterr.* **2022**, *61*, 563–599. [\[CrossRef\]](#)
81. Crespo, M.; Guirado-Manzano, L.; Sarmiento, D.; Guirado, E.; de Vicente, A.; Fernández-Ortuño, D.; Arrebola, E. Fungal species of *Botryosphaeriaceae* associated with avocado dieback in Southern Spain. *Plant Pathol.* **2026**, *75*, e70079. [\[CrossRef\]](#)

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.